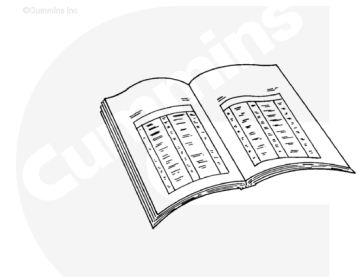


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



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⚠ WARNING ⚠

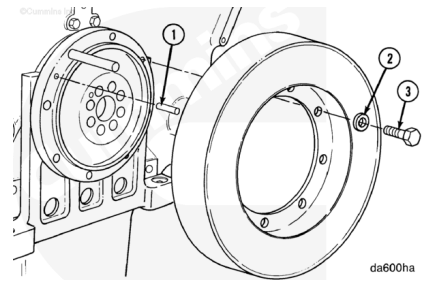
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>)
- Disconnect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/28/28-012-022-tr-gc.html>)
- Remove the front power take-off (PTO) stub. Refer to Procedure 009-021 in Section 9. (</qs3/pubsys2/xml/en/procedures/28/28-009-021-tr-gc.html>)
- Remove the crankshaft adapter. Refer to Procedure 001-017 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-017-tr-gc.html>)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not pry on the vibration damper with tools that are sharp or pointed, damage to the damper can result.

Note : The roll pin (1) does not have to be removed unless it is damaged.

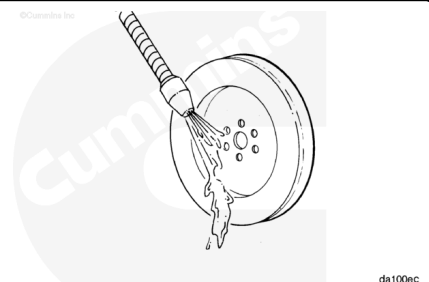
Note : The use of a guide bolt aids the disassembly procedure.

Remove the Society Automotive Engineers (SAE) Grade 8 capscrew (3), the hardened washers (2), and the vibration damper.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

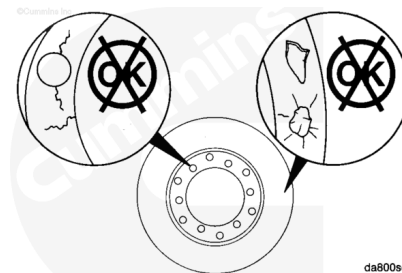
Note : Vibration dampers have a limited service life. The damper **must** be inspected as per the engine maintenance guidelines. Reference the K38, K50, QSK38 and QSK50 Operation and Maintenance manual. The vibration damper **must** be replaced during an end of life rebuild.

Note : Do not repair or balance a viscous damper in the field.

Use solvent to clean the exterior of the damper. Dry with compressed air.

Inspect the mounting flange for cracks. Inspect the housing for dents, bulges, or leaks.

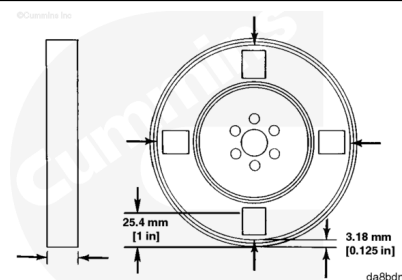
If the part is damaged, it **must** be replaced.



Use a paint solvent and a fine emery cloth to remove the paint from the front and the back of the housing at the four areas as shown.

Measure and record the thickness at two points at each of the four locations:

- Measure the thickness at 3.18 mm [0.125 in] from the outer lip.
- Measure the thickness at 25.4 mm [1.0 in] from the outer lip.



If the variations between any of the eight measurements exceed 0.25 mm [0.010 in], or if the thickness at any point exceeds the maximum thickness dimensions indicated in the following table, replace the damper.

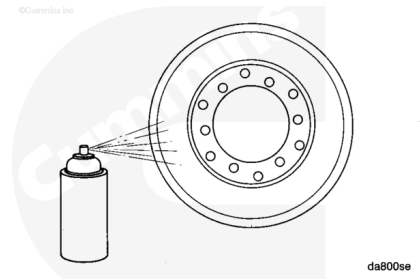
Maximum Vibration Damper Thickness		
mm		in
67.92	MAX	2.674

Note : If the damper has been in service for 24,000 hours or more, it **must** be replaced, regardless of the thickness measurement.

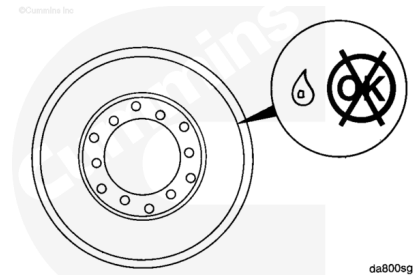
The next three graphics illustrate the procedure to check the damper for fluid leakage.

- Use the crack detection developer, Part Number 3375434, or equivalent. Spray the rolled lip of the damper.

Note : The crack detection kit, Part Number 3375432, contains the necessary cleaner, the penetrant, and the developer to check for cracks using the dye penetrant method.



- Check for fluid leakage around the rolled lip. The damper **must** be replaced if there is any fluid leakage.



Damper Replacement Recommendations

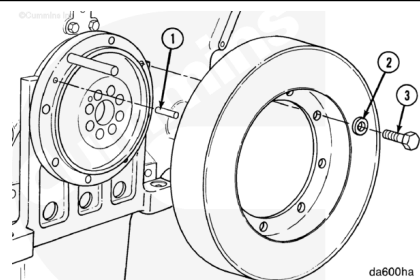
- Dampers that have exceeded their useful life will **not** function properly and can cause engine damage due to excessive torsional vibration in the engine.
- Viscous vibration dampers **must** be replaced during an end of life rebuild.

Note : Always replace the vibration damper after a crankshaft nose failure. Crankshaft nose breakages are primarily caused by excessive torsional activity.

Install

⚠ WARNING ⚠

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⚠ WARNING ⚠

The capscrews (2) must be SAE Grade 8. The washers (3) must be hardened washers. If the wrong parts are installed, the capscrews can fail, allowing the damper to become loose and cause personal injury.

⚠ CAUTION ⚠

Dampers are available with standard and high viscosity fluid inside the damper. If the damper has been replaced, check to be sure the correct damper is installed.

⚠ CAUTION ⚠

The vibration damper must be aligned correctly with the crankshaft adapter or the valve and injector adjustment marks on the vibration damper will be in the wrong position.

Note : K38 and K50 vibration dampers are not the same. The valve and injector adjustment marks stamped on the outside diameter of the damper are different. Check to be sure the correct part is installed.

If necessary, install the roll pin (1) in the crankshaft adapter.

Check the mating surfaces of the damper and the adapter to be sure they are clean, dry, and free of nicks or burrs.

Align the hole for the pin in the damper with the roll pin.

Position the damper on the adapter.

Use clean engine oil to lubricate the capscrews and washers.

Install the washers and capscrews.

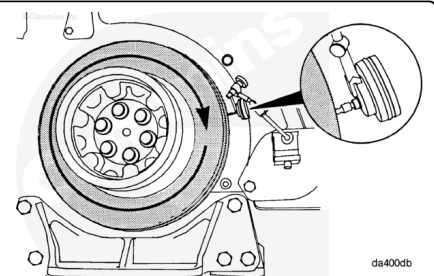
Torque Value: 190 n·m [140 ft-lb]

Eccentricity Check

Use a dial indicator.

Note : If the engine has two dampers, measure the eccentricity of both.

When setting up the indicator, make sure the indicator tip does **not** touch the valve and injector adjustment marks as the crankshaft is rotated.



Adjust the indicator, as shown, to measure the radial alignment of the damper(s). Turn the crankshaft 360 degrees, and record the total runout.

Vibration Damper Eccentricity (Total Indicator Reading)		
mm		in
0.51	MAX	0.020

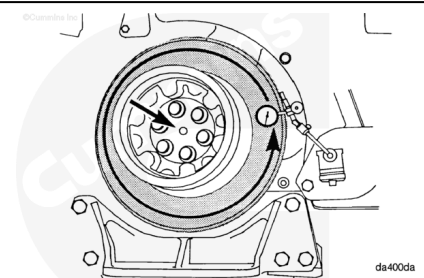
If the measurement is **not** within specifications, make sure the vibration damper pilot is aligned properly.

Wobble Check

Note : Keep the crankshaft at one end of its thrust when damper wobble is being measured.

Place the indicator plunger tip 13 mm [0.50 in] from the outside circumference on the face of the damper. Rotate the crankshaft 60 degrees and record the total reading.

Vibration Damper Face Runout (Total Indicator Reading)		
mm		in
0.58	MAX	0.023



Note : When the dial indicator tip is positioned more than 13 mm [0.50 in] from the circumference, the wobble specification is different. To determine the maximum face wobble when the indicator tip is in another location:

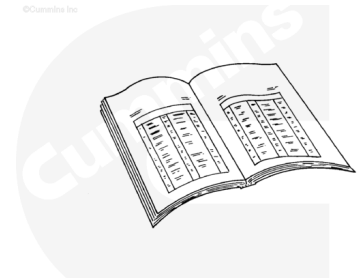
- Measure the distance from the center of the crankshaft to the point the indicator tip touched the face of the damper, then multiply the distance by 0.064 mm [0.0025 in] to obtain the maximum total indicator runout allowed.

If the measurement is **not** within specifications, check for foreign material between the crankshaft and the adapter (or pulley), and between the adapter (or pulley) and the vibration damper.

Finishing Steps

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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the crankshaft pulley. Refer to Procedure 001-022 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-022-tr.html>)
- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Install the crankshaft adapter. Refer to Procedure 001-017 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-017-tr-gc.html>)
- Install the front PTO stub. Refer to Procedure 009-021 in Section 9. (</qs3/pubsys2/xml/en/procedures/28/28-009-021-tr-gc.html>)
- Connect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/28/28-012-022-tr-gc.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine and check for proper operation.

